

Core rc16 memory layout

Foenix F256 K2 + Jr2 | Current RTL and NitrOS-9 max-RAM kernel | 14 September 2026

1,792 KB

CPU RAM pool with FLASHDIS = 1

224 x 8 KB

MMU blocks available to the max-RAM kernel

2,048 KB

Physical SRAM capacity: 1M x 16 bits

01 The 256-entry MMU block space

Byte ranges below are block x \$2000 through block x \$2000 + \$1FFF. Device selections are not SRAM accesses.

MMU BLOCKS	BYTE ADDRESS RANGE	SPAN	RESET: FLASHDIS = 0	MAX-RAM: FLASHDIS = 1
\$00-\$3F	\$000000-\$07FFFF	512 KB	SRAM	SRAM / 64 blocks
\$40-\$7F	\$080000-\$0FFFFFF	512 KB	Flash select	SRAM / 64 blocks
\$80-\$9F	\$100000-\$13FFFF	256 KB	Cartridge / expansion select	SRAM / 32 blocks
\$A0-\$BF	\$140000-\$17FFFF	256 KB	SRAM	SRAM / 32 blocks
\$C0-\$C7	\$180000-\$18FFFF	64 KB	Sectored device space	Device space / NotRAM
\$C8-\$CF	\$190000-\$19FFFF	64 KB	No RAM selection	NotRAM
\$D0-\$EF	\$1A0000-\$1DFFFF	256 KB	SRAM	SRAM / 32 blocks
\$F0-\$FF	\$1E0000-\$1FFFFFF	128 KB	No RAM selection	NotRAM

The kernel excludes \$C0-\$CF and \$F0-\$FF. The 1,792 KB total is the RAM pool before OS, screen and application allocations.

02 Address translation and the rc16 control bit

The CPU selects a block through an MMU slot. VICKY and DMA use SRAM byte addresses directly.

CPU -> MMU -> SRAM

byte address = (block x \$2000) + 13-bit offset

Example: \$D0:\$0000 -> \$1A0000 -> word pins \$D0000

Block \$D0 ends at \$1A1FFF; the whole \$D0-\$EF range ends at \$1DFFFF.

MMU_IO_CTRL: CPU \$FFA1

Bit 2: FLASHDIS. Reset = 0; set to 1 for RAM in \$40-\$9F.

Read bit 7 = 1 identifies support. Preserve the other control bits.

Bits 0 / 1 enable fixed internal RAM at \$FD00 / vectors at \$FFF0.

SRAM pins carry a WORD address: byte address >> 1; the low byte-address bit selects a byte lane. No +\$30 or address folding.

03 Device pages and fixed CPU windows

These are peripheral views, not extra RAM. Mapping a device page into a CPU slot selects its page-relative offsets.

MMU PAGE	PAGE OFFSETS	CONTENTS	CPU ADDRESS	FIXED WINDOW / CONTROL
\$C0	\$0000-\$0FFF	Gamma tables / mouse image	\$FD00-\$FDFF	256-byte internal RAM when \$FFA1 bit 0 = 1
\$C0	\$1000-\$12FF	Bitmap, tile and misc controls	\$FE00-\$FEFF	System, IRQ, timers, serial, mouse, DMA, math
\$C0	\$1300-\$16FF	128 sprite attribute records	\$FF00-\$FF5F	SD, splash SPI, WizFi, MIDI, W6100, VS1053
\$C0	\$1700-\$177F	Text foreground/background LUTs	\$FF90	DIP-switch input
\$C1	\$0000-\$0FFF	Two font banks	\$FFA0	Active LUT bits 1:0; edit LUT bits 5:4
\$C1	\$1000-\$1FFF	Four graphics palette LUTs	\$FFA1 / \$FFA8-\$FFAF	I/O control / eight MMU slot registers
\$C2 / \$C3	Page-relative offsets	Text characters / text attributes	\$FFC0-\$FFCF	Video master, layers, border and bitmap mode
\$C4	Sound register offsets	SID / PSG register space	\$FFF0-\$FFFF	16-byte vector RAM when \$FFA1 bit 1 = 1

Selected device resources are shown above; gaps in those tables are not a promise of usable storage or a connected peripheral.

HOW TO READ THIS MAP

RAM mode: the rc16 max-RAM kernel detects \$FFA1 bit 7 and sets FLASHDIS. A core alone does not enlarge the kernel block pool.

Video / DMA: physical SRAM addressing bypasses the CPU LUT and FLASHDIS. The CPU device-space holes do not remove SRAM capacity.

Fixed windows override the selected CPU slot. rc16 inhibits SRAM writes at \$FFA0-\$FFAF so MMU register writes cannot leak into RAM.

SOURCE BASIS

TyVKy2K2x1_MMU_Register.v | TyVKy2K2turbo_MMU_FNX6809.v | defs/wildbits.d | wb/max_ram_upgrade: krnp2.asm

Original chart artwork and wording. Reference image used only for the colorful tabular format and page proportions.